

Types Without Taxa

A Covariance-Matched-Null Multiverse Test of Categorical versus Continuous Personality Structure

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Abstract

Run a clustering method on personality data and it will hand you types, whether or not any types are there. That makes a familiar announcement, that a dataset contains some particular number of personality types, hard to take at face value; the number might be a fact about people, or only a fact about the method. This paper offers a way to tell the two apart, and tries it on the personality case. I compare each of three large public datasets (about a million respondents, across the IPIP-NEO-120, an IPIP Big-Five inventory, and the IPIP-HEXACO) to a matched null: a synthetic population built to copy the data's trait distributions and full correlation structure exactly, but to contain no types by construction. That comparison runs across all fourteen covariance assumptions a standard mixture model can make, and a taxometric test adds a second, independent reading. Everything was preregistered and run on held-out data. The number of types found ranges from one to ten depending only on the covariance assumption; by the preregistered measure it never exceeds what the typeless null produces; and no trait shows a categorical signature. Real data does depart a little from the null, but not by gaining types or turning categorical. The contribution is not a verdict on personality; trait structure has long been argued to be dimensional. What is new is the test, which any future claim of N types can be held to.

Keywords: personality structure; categorical versus continuous; mixture models; matched null; taxometrics; multiverse analysis; preregistration

Introduction

Few ideas about personality are as easy to like as the idea that people come in kinds. We call each other introverts and extraverts as though naming two species; we sort ourselves into four-letter codes; and every few years a study announces that the data, left to themselves, fall into a handful of personality types. One widely cited paper found four (Gerlach et al., 2018). The appeal is no mystery. A type is something you can hold in your head; a position in a five- or six-dimensional space is not.

But a clustering method will find types whether or not any are there. Give a mixture model a single smooth cloud of data and it still reports two or three or ten groups, because each added group trims the fit a little and the selection rule takes the trade. A clean partition is not much of a finding by itself. The question worth asking is not how many types a method returned, but whether they do anything more than trace the shape the data already had.

On personality itself, the argument is largely over. Taxometric and structural studies have looked for categories in trait data for years and kept finding gradients instead (Haslam et al., 2012; Haslam et al., 2020). I am not trying to reopen that. What the settled view still lacks is a way to check a specific claim, this dataset and these respondents and this many types, against a null that means something, and a sense of how much the claim depends on choices made before any result is in view.

Past attempts to impose that discipline fall short in ways worth naming. The best-known typology study tested its clusters against data with all the associations shuffled out (Gerlach et al., 2018), but data with no correlations is a low bar, and almost anything real clears it. A later comment said as

much, without building the alternative it asked for (Katahira et al., 2020). Taxometrics asks the categorical-versus-continuous question head-on (Meehl, 1995; Ruscio et al., 2006), but it works from a fixed set of indicators and has nothing to say about how many clusters there are, or how that number moves when the model changes.

My approach has three parts, and none is new on its own. The null is a Gaussian copula. It keeps every trait's distribution and the whole covariance matrix and scrambles only what is left, the higher-order dependence, so a cluster counts as real only if it survives a world that already shares the data's margins and correlations. This is the constructed null the earlier comment called for. I then refuse to report a single number; the same test runs across all fourteen covariance parameterizations of the mixture model, and I show the whole spread of answers rather than the one I happen to prefer. And I bring in taxometrics as an outside witness, so the verdict does not rest on one method's assumptions. All of it is aimed at the personality-types question, across three datasets.

The picture that comes back is consistent. Change the covariance assumption and the number of types walks from one to ten. Hold the rest fixed and, by the measure I committed to in advance, that number is no different from what the typeless null gives. No trait crosses the taxometric line. None of that is new about personality. What it shows is that the test works, and that the next typology has something to answer to.

Background

A cluster solution settles nothing on its own

The wish to sort people into kinds is old, and it keeps coming back in quantitative dress: clinical subtypes, learner profiles, data-driven personality types. The trouble is always the same. Clustering and mixture methods exist to partition; give them continuous data and they partition it, and a selection criterion will almost always prefer more than one piece. So “the data support k clusters” quietly mixes a fact about people with a fact about the procedure, and the procedure rarely has to answer for itself, to say what a world with no types would have produced under the same treatment.

What is already settled

Whether traits come in classes or in degrees has been asked many times and answered the same way, in degrees (Haslam et al., 2012; Haslam et al., 2020). Because that much is settled, the thing worth adding is not the conclusion but a tool: something that turns a particular cluster-count claim into a claim that can fail, and shows how much it leans on choices no one inspects.

The null is where the argument lives

An apparent grouping is “real” only against a null that says what real means here. Shuffle every association out of the data (Gerlach et al., 2018) and you have set the bar on the floor; the correlations alone will clear it. Katahira et al. (2020) saw this but stopped at the diagnosis. The cure is borrowed from elsewhere, from the surrogate-data and matched-null traditions of nonlinear time series, network neuroscience, and ecology, where you fix everything innocent about a dataset and randomize

only the thing in question. For a claim about clusters the copula null is the natural choice: keep the margins, keep the covariance, and let only higher-order dependence vary, so any extra clustering has to come from structure the null cannot hold.

A second opinion

Taxometrics reaches the same question by a different road, asking whether the covariation among a construct's indicators looks more like one dimension or two latent groups (Meehl, 1995; Ruscio et al., 2006). The Comparison Curve Fit Index scores the observed curves against curves simulated from frankly dimensional and frankly categorical data. Because its assumptions are not a mixture model's, it makes a useful outside check, though it works on a fixed indicator set and says nothing about how many clusters there are.

Reporting the spread, not a point

That a result can hang on a defensible choice is the premise of multiverse and specification-curve analysis (Steegen et al., 2016; Simonsohn et al., 2020), with a longer root in the sensitivity arguments of econometrics (Leamer, 1983). The honest move is to run the reasonable specifications and show the distribution instead of defending one. For the number of types, the choice that matters is the mixture model's covariance assumption, which I vary in full; the matched null then says where, on that spread, a typeless world would have landed.

Method

Transparency

Everything below was fixed in advance. Before I opened the held-out data I registered the decision rule and the specification grid (OSF, seed 20260620), and what follows reports every dataset, every covariance model, and the whole rule, with nothing added or dropped once the hold-out was in view.

Data

I reanalyze three large public datasets, each already scored to traits and, where the instrument allows, facets: the IPIP-NEO-120 (Johnson, 2014) (410,376 respondents; five domains, thirty facets); an IPIP fifty-item Big-Five set (Goldberg, 1999) (603,322; five domains, no facets defined); and the IPIP-HEXACO (22,734; six domains, twenty-four facets). Three instruments rather than one, so that a verdict has to repeat across measurement systems instead of surviving in a single one.

I split each dataset in half (seed 20260620) into an exploratory and a confirmatory part, and sealed the confirmatory part before fitting anything reported here. For the NEO-120 and the fifty-item set the held-out respondents are almost all unseen, about 95 to 97 percent; HEXACO is small enough that the halves overlap, with about 88 percent of its hold-out also in the exploratory split, so I treat it as a robustness check rather than a clean confirmation. Every model was fit to a random subsample of the held-out part, 8,000 cases at the domain level and 5,000 at the facet level. One housekeeping point matters later: mixtures tend to select more components as the sample grows, so I always compare the real data and its null at the same sample size.

The covariance multiverse

I model each dataset as a Gaussian mixture and let “types” mean the components a BIC selection keeps. `mclust` (Scrucca et al., 2016) offers fourteen ways to constrain a component’s covariance, in volume, shape, and orientation, running from the spherical (EII, VII) through the diagonal (EEI, VEI, EVI, VVI) to the fully ellipsoidal (EEE through VVV). For each, I fit one to ten components and record the number BIC prefers. The fourteen answers, taken together, are the specification curve: how many types you find as a function of an assumption that carries no meaning about people.

The Gaussian-copula matched null

To ask whether a clustering is more than the data’s own margins and covariance, I compare it to a null that holds both fixed and frees only the rest. From a real $n \times p$ subsample I take the correlation matrix and its Cholesky factor, draw a matrix of independent standard-normal numbers, correlate them through that factor, and then, column by column, replace each correlated normal variable with the sorted real values of that variable laid down in the same rank order. What comes out has every marginal exactly, the full covariance to within sampling error (the largest correlation it misses is about 0.009), Gaussian dependence, and no clustering built in. That is the right null for the question. It fixes everything a single-cloud description already carries, so any excess clustering in the real data is structure the null could not have. I drew sixty such nulls for each domain dataset and fifty for each facet dataset.

Taxometric analysis

For the categorical-versus-continuous question head-on, I hand each trait's facets (six per NEO-120 trait, four per HEXACO trait) to the two taxometric procedures named in the registration, MAMBAC and MAXEIG, through RTaxometrics (Ruscio, 2023). The CCFI scores the observed curves against curves simulated from explicitly dimensional and explicitly categorical comparison data, returning a number between zero and one: below .45 reads dimensional, above .55 categorical, with an ambiguous band between (Ruscio et al., 2006). I report each trait's CCFI as the average of the two and keep the spread between them in view; the L-Mode curve, which the routine also returns, agreed throughout. The fifty-item set has no facets, so this part covers only the NEO-120 and HEXACO.

The pre-registered decision rule

The rule was set before the hold-out opened; the full text is in the registration. In short: I credit an instrument with real structure only if its clustering gain, the BIC improvement of the best mixture over a single component, beats the 95th percentile of the matched null's. I call the structure continuous only if four things hold together: mean within-trait CCFI under .45 with no trait at .55 or above, in both instrument families; fewer than half of respondents confidently placed in any one component, at a posterior above .8; the first split, from one component to two, carrying more than three-quarters of the total BIC gain; and the spread of selected counts across the fourteen models indistinguishable from the null's, with the real median inside the null's 95 percent interval. The opposite verdict, real types, needs the opposite signs: a CCFI at .55 or above that repeats across instruments, a clear BIC minimum with gaps between component centers, most respondents con-

fidently placed, and a count that holds steady across datasets. The NEO-120 and the fifty-item set are the confirmatory pair and have to agree; HEXACO is robustness only.

For the record, the statistics are these. ΔBIC is the best-of-fourteen BIC minus the one-component BIC; the count summarized across models is the median of the fourteen; the first-split share is the two-component gain over the total gain, both measured on the best-of-fourteen envelope; and the confident-assignment share is the fraction of respondents whose top posterior exceeds .8 under the selected solution.

Software

R (Team, 2024), with `mclust` for the mixtures and `RTaxometrics` for the CCFI; seed 20260620 throughout. Data, code, and the registration are public.

Results

The covariance assumption sets the count

Start with the count itself, and it refuses to settle. Across the fourteen covariance models the number of components BIC keeps runs from one to ten (Figure 1). At the facet level it spans the whole grid in both the NEO-120 and HEXACO; even at the domain level it runs from three to ten in HEXACO and five to ten in the NEO-120. The parameterization means nothing about people, it is a convenience, yet it moves the apparent number of types across almost the entire range anyone might report. The drift has a direction. The richer ellipsoidal models keep fewer components than the plain spherical and diagonal ones, because a flexible covariance can absorb the continuous shape that a rigid one

can only chase by adding pieces. So the count isn't telling you about the respondents. The model settled it before anyone looked at the data.

There is real structure, but weak against the data's own shape

A wandering count does not by itself prove the structure is empty, so I compared each dataset to the matched null. In all five slices the real clustering gain beat the null's 95th percentile: 717 against 485 for the NEO-120 domains, 1,369 against 910 for the fifty-item domains, 816 against 215 for the HEXACO domains, and 4,013 against 1,073 and 2,701 against 337 for the two facet sets. Real personality data, then, is not just its margins and covariance dressed up; some genuine higher-order structure is there in every case. But this mostly certifies that the data are not Gaussian, which was never in doubt; the null fixes only the linear part of the dependence, so beating it is a low bar. The two comparisons that follow, the count and the taxometric curves, are the ones that carry the argument.

The count is no larger than a typeless world's

The next question is whether that real structure shows up as more types. The null has no types by construction, but it still produces multi-component solutions of its own, so the test is whether the real count runs higher. By the measure I registered, the median count across the fourteen models, it does not. The real median sits inside the null's 95 percent interval in every slice: 5 in [4.0, 7.0] for the NEO-120 domains, 4 in [3.0, 6.4] for its facets, 8 in [6.0, 8.8] for the fifty-item domains, 7 in [3.5, 7.5] for the HEXACO domains, and 4 in [1.0, 4.9] for its facets. The grey band in Figure 1 marks the null's interval; wherever the dashed real median falls inside it, the typical number of

The typical number of 'types' matches a typeless null (pre-registered median statistic)

Each dot = one of 14 covariance models on held-out data (matched n). Grey band = 95% interval of the matched-null MEDIAN count; dashed line = the real median count. Real line inside the band means the typical count is indistinguishable from data with no types.

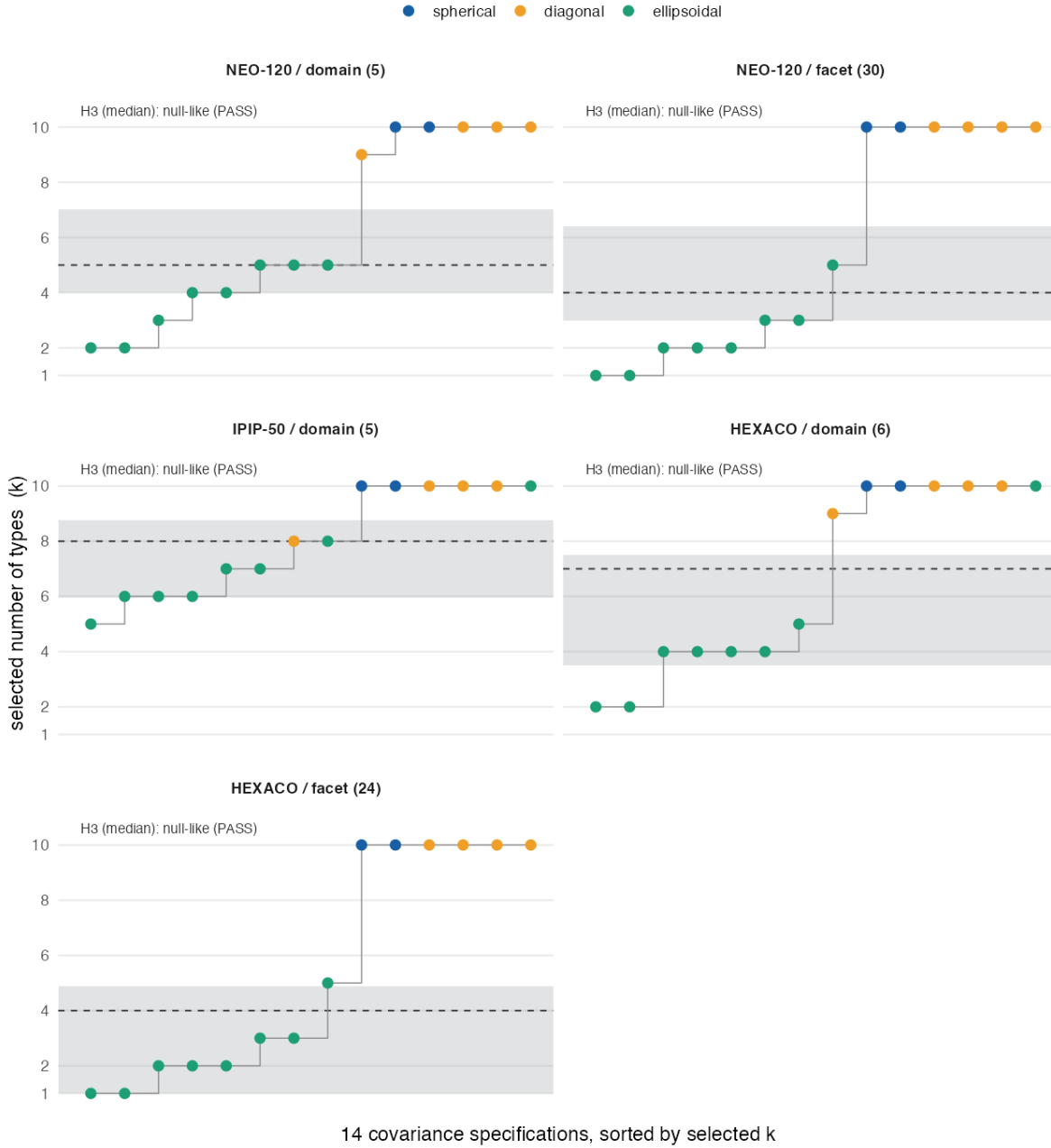


Figure 1: The BIC-selected number of mixture components (“types”) across fourteen covariance parameterizations, on held-out data (matched sample size). Each point is one parameterization, coloured by covariance family and sorted by selected count. The grey band marks the 95% interval of the matched-null median count; the dashed line is the real median. A real line inside the band means the typical count is indistinguishable from data with no latent types.

types is one the typeless world could have produced.

The mean tells a slightly different story, and the difference is worth keeping. Averaged rather than taken at the median, the count does exceed the null in the fifty-item and HEXACO sets. The two part ways because the per-model counts are skewed: a few covariance models run all the way to the ten-component ceiling, and that tail pulls the mean up while leaving the median where it is. I registered the median, and I report the mean too. That a verdict can hinge on which summary of the same fourteen numbers you take is, if anything, this paper's own argument turned on itself, and a reason to fix the statistic before looking.

No trait looks like a category

The taxometric test points the same way. No trait reached the categorical threshold in either instrument (Figure 2): across eleven traits the highest was NEO-120 Openness, at .535. The NEO-120 facets averaged .375, the HEXACO facets .430, the latter just below the dimensional cutoff. A few traits sat in the ambiguous band, the highest NEO-120 Openness (.535) and HEXACO Honesty-Humility (.520), but none crossed into categorical territory. One procedure occasionally reached higher, which Figure 2 shows rather than hides; the registered figure, the average of the two, did not.

People do not sort cleanly, and one cut does most of the work

Two more checks, both registered, lean the same way. Under the selected solution, fewer than half of respondents could be placed in any single component with much confidence, at a posterior above .8: 14.6 and 48.0 percent for the NEO-120 domains and facets, 25.1 percent for the fifty-item set, 30.0

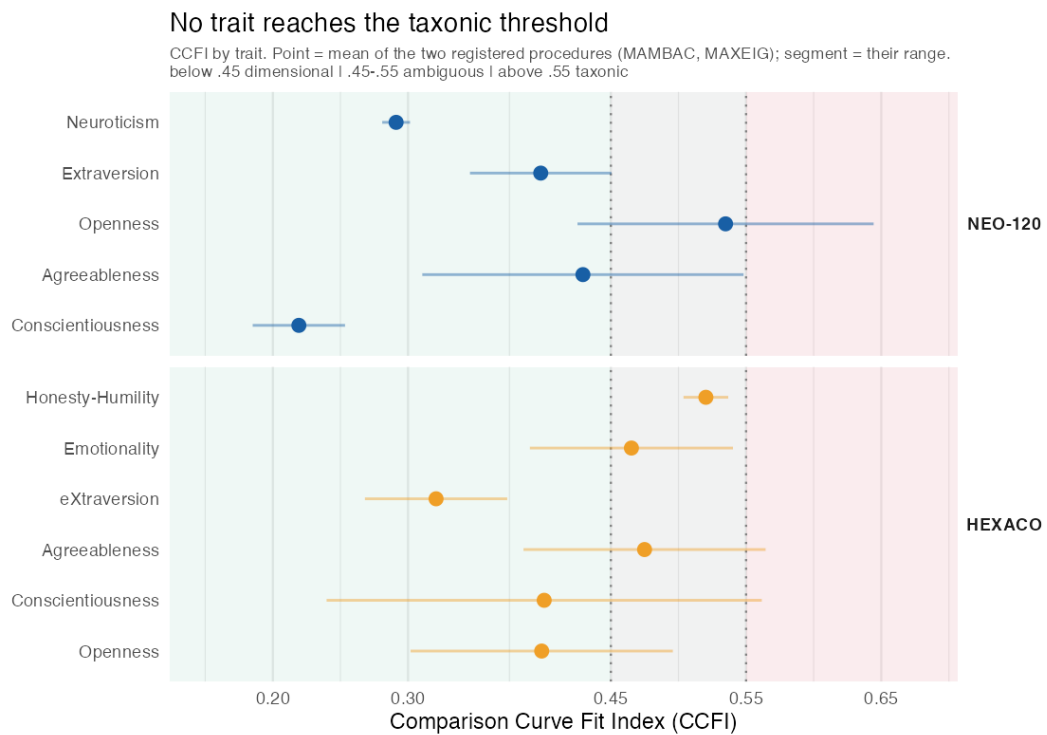


Figure 2: CCFI by trait. Each point is the mean of the two registered procedures, MAMBAC and MAXEIG; the segment is their range. Values below .45 read dimensional, .45 to .55 ambiguous, above .55 taxonic. No trait's mean reaches the taxonic threshold.

and 46.0 percent for HEXACO. And the first cut, from one component to two, carried more than three-quarters of the total BIC gain in four of the five slices, 88.7 and 89.6 percent for the NEO-120 and 82.8 and 98.9 percent for HEXACO, the lone exception being the fifty-item domains at 61.8 percent. Where the model did split people, it found them spread along the splits, not gathered on one side.

The test can find types when they are there

A failure to find types is worth only as much as the test's power to find them, so I checked that power directly, on synthetic data with a known structure (Figure 3). I built a mixture whose two components share identical marginal distributions and differ only in the orientation of their dependence, so the clustering lives in the joint structure and nowhere in the margins. The test caught it: as the signal grew the real median count climbed out of the null, from one at no signal to nine at a strong one, while the matched null stayed at one throughout. A no-signal version was read, correctly, as typeless. So the test is not a machine that only ever returns "continuous"; when separated types are present in the dependence it fires, and fires harder the clearer they are.

It also has a blind spot worth stating plainly. When I instead built well-separated clusters whose separation was large enough to register in the univariate distributions, the test went quiet: the matched null, which copies those distributions exactly, then produced the same clustering on its own. That is the cost of holding the margins fixed, and here it is the right cost. A type already visible in each variable's histogram is one a continuous distribution with those histograms also produces, so the test declines to call it a discovery. Personality trait distributions are unimodal, so the test operates in the first regime, not the second.

Positive controls: when the test fires, and when it stays quiet

Synthetic data with known types. Grey band = matched-null median-k (95%); points = real median-k, red when it exceeds the null.
A: types living only in the dependence are detected, monotonically. B: types so separated they enter the margins are reproduced by the null (test is conservative).

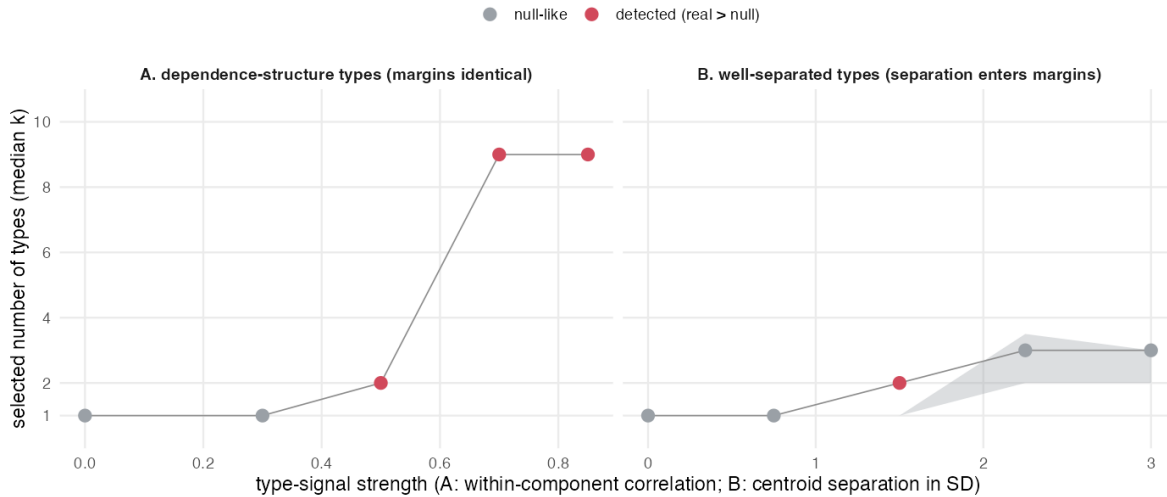


Figure 3: Positive controls on synthetic data with known structure. Grey band: matched-null median count (95%). Points: real median count, red where it exceeds the null. Left, types that live only in the dependence (identical margins, opposite correlation orientation) are detected, and detection grows with the signal. Right, types so separated that the separation enters the margins are not flagged, because the margin-preserving null reproduces them; for unimodal personality data the test operates in the left regime.

The verdict

Held against the rule I set in advance, the categorical signs are missing everywhere: no CCFI at .55 that repeats, no clear BIC minimum, no count that holds still. The NEO-120, the dataset with a clean hold-out, meets all four continuity conditions, and so does HEXACO, whose mean CCFI (.430) sits below the dimensional cutoff. The fifty-item set meets three of the four; it misses only the first-split threshold, at 61.8 percent against 75, and has no facets to give it a CCFI. None of these is a categorical sign. So I find no evidence for separated types in any of the three. The clusters they yield are ones the matched null reproduces, and the mild real structure that remains never turns into extra types or into categories. HEXACO I keep as a robustness check, since its hold-out is not clean.

Discussion

I set out to ask whether the types a mixture model reports in personality data are categories of people or just the data's continuous shape, and I tried to settle it with a test rather than one more clustering. Each dataset met a copula matched null that keeps its margins and covariance; the test ran across all fourteen covariance models; and a taxometric index gave a second reading. Across three large datasets the count ran from one to ten on the covariance assumption alone, sat inside the typeless null by the registered measure in every case, and never reached the taxometric line for a category.

I take this as a result about method, not about people. That traits are continuous is old news (Haslam et al., 2012; Haslam et al., 2020); I am not adding to that case so much as handing it an instrument, one that makes the standing claim “this dataset has N types” something that can be checked, and shows the number at its heart to be set by a choice that means nothing.

The claim is meant to be narrow. The null tests one thing, whether the clustering beats the linear dependence already in the data, and on that test the real data does carry structure the null cannot make; every dataset cleared the bar. But that leftover structure does not become more types, and it does not become taxonic curves. So the honest line is “no evidence for separated types,” not “perfectly continuous.” Something real is there, in every dataset, below the level of a category, and I do not try to name it here. Skew, a little nonlinearity, some unevenness in density: any of these would do, and none is a kind of person.

One thing in my own analysis makes the point better than I could. Whether the count beats the null depends on whether you read it at the mean or the median, because a few models run to the ceiling

and skew the average. I registered the median; the mean says otherwise; I report both. A verdict that turns on which summary you pick is exactly the trouble this paper is about, and a reason to name the statistic before the data are in front of you.

The test is not about personality in particular. Any claim that continuous measurements fall into so many groups can be put to a matched null, and the habit of reporting a spread of answers instead of a favorite one travels anywhere a modeling choice is beside the point. It sits alongside taxometrics, which asks the same question from other assumptions, and it builds the constructed null that earlier critics of typology work asked for and did not supply (Katahira et al., 2020); an independence null, again, asks too little.

There are limits. A multiverse is only as honest as the choices it spans, and I varied the covariance assumption fully while holding the mixture family, the selection criterion, and the resolution fixed; a wider one could be drawn. The null tests linear dependence, so a clustering driven by something genuinely curved but still continuous would read as real without being a category, which is close to what I see, and which the null is right not to call a type. The datasets are large but self-selected, three of a kind, all from the IPIP pool, and subsampled to keep the computation in hand; HEXACO's hold-out is not clean. And I test for separated categories and their number, not for whether a rough typology is ever a useful way to talk; finding no taxa does not mean a typology can never be a convenient shorthand, only that it is not a discovery of kinds.

The use of all this is small and practical. A number of clusters is not yet a finding. It becomes one only when it beats what the data's own margins and covariance produce, and survives the choices, here the covariance assumption, that should have made no difference. I have left the test where others can pick it up, so that the next claim of N types can be asked, first of all, whether N belongs

to people or to the procedure.

Conclusion

The number of types a method reports is not, by itself, a fact about people. For personality, a case studied to the point of consensus, I have shown the count drifting across the whole range as one inconsequential assumption is changed, staying inside what a typeless null produces, and never showing a categorical signature in any trait. None of this is news about personality; the point is the test. It can make a claim of N types fail against a null that means something, and it shows how much the claim leaned on choices nobody was watching. The materials are public, so the next claim can face the same check. Whether a structure is real and whether it is a set of kinds are two questions, and a count of clusters answers neither until it is asked against the right null.

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